

Onboard SAR Processing for Ship Detection: Modeling and Evaluation

Date: 2026-08-24

Presented by: Jen-Tse Chen

Advisor: Pei-Yun Tsai

Outline

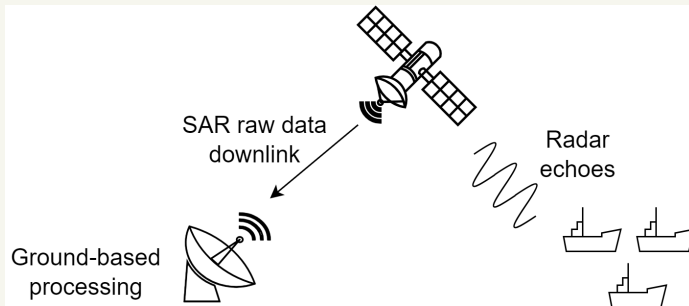
- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.



Problem and Motivation

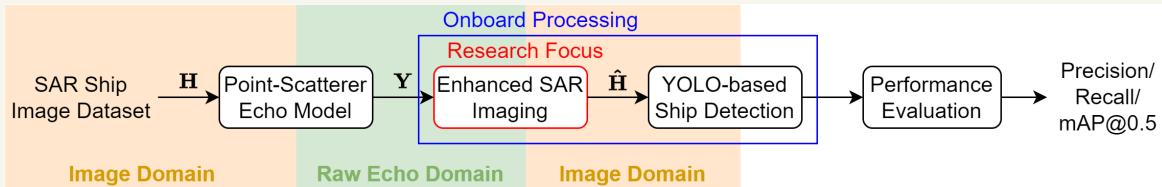
- ▶ Ground-Based Processing
 - ▶ Large raw-data volume
 - ▶ Limited downlink bandwidth
 - ▶ Delayed target information
- ▶ Onboard Processing
 - ▶ Reduced data transmission and detection latency
 - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
 - ▶ Lightweight SAR processing for low-latency ship detection

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection



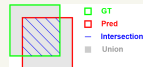
► Evaluation Metrics [1]

- Precision: $\frac{TP}{TP+FP}$, Recall: $\frac{TP}{TP+FN}$

TP: True Positive, FP: False Positive, FN: False Negative

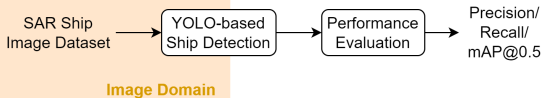
- mAP@0.5: Mean AP across classes at IoU threshold = 0.5

IoU = overlap area / union area of predicted and ground-truth boxes



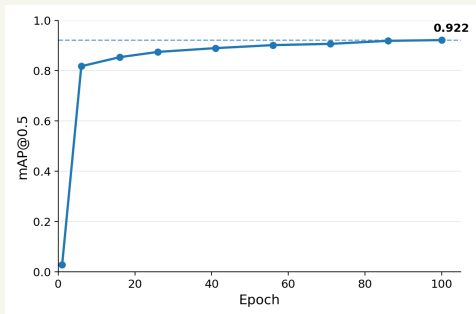
Dataset and Detection Model Setup

- ▶ HRSID [2]
 - ▶ High-Resolution SAR Images Dataset
 - ▶ Training: 3,642 images
 - ▶ Validation: 1,962 images
 - ▶ Image size: 800×800 pixels
- ▶ YOLO Detection Model
 - ▶ YOLO11m fine-tuned on HRSID (100 epochs)



Validation Performance

- ▶ Precision: 0.910
- ▶ Recall: 0.840
- ▶ mAP@0.5: 0.922

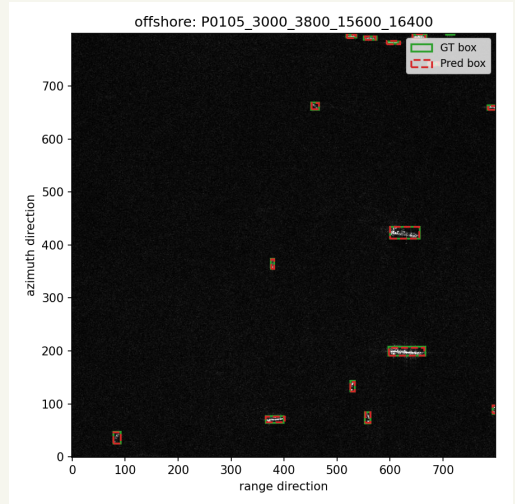
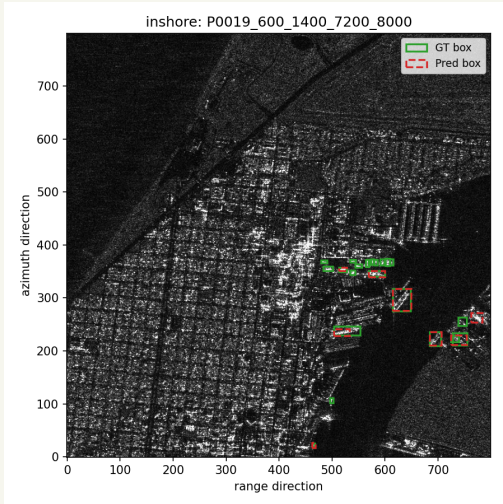


YOLO Ship Detection Results (1)

	Overall	Inshore	Offshore
Images	1,962	369	1,593
Precision	0.913	0.803	0.982
Recall	0.846	0.741	0.965
mAP@0.5	0.918	0.817	0.984

- Offshore scenes achieve better detection performance than inshore scenes [3].

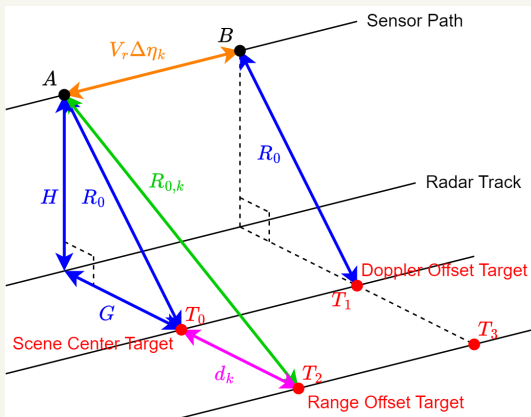
YOLO Ship Detection Results (2)



Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range $R_k(\eta_m)$.
- $$R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta\eta_k)^2 V_r^2}$$
 - $R_{0,k} \approx R_0 + d_k$
 - R_0 : slant range to scene center
 - d_k : ground-range offset of scatterer k , approximated as slant-range offset
 - $\eta_m = m\Delta\eta$: slow time
 - $\Delta\eta_k$: azimuth-time offset of scatterer T_k

Point-Scatterer Echo Model: Signal Synthesis

► Geometry-Dependent Delay and Phase

$$\text{► } \tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}, \phi_k(\eta_m) = -4\pi \frac{R_k(\eta_m)}{\lambda}$$

► Echo Synthesis: $\mathbf{S} = [s_{m,n}]_{M \times N}$

$$s_{m,n} = \sum_k A_k \text{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r (\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

► $\eta_m = m\Delta\eta$: slow time, $\tau_n = n\Delta\tau$: fast time

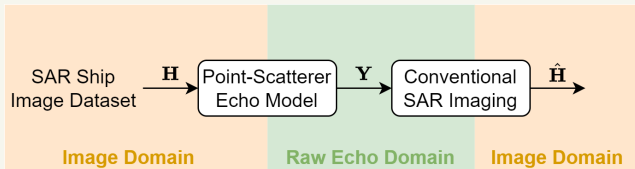
► A_k : scattering coefficient of scatterer k , $K_r = \frac{B_r}{T_r}$

K_r : chirp rate, B_r : chirp bandwidth (Hz), T_r : pulse duration (sec)

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Conventional SAR Imaging



- Imaging goal: Recover $\mathbf{H}$ from $\mathbf{Y}$.
- Conventional methods: RDA, CSA, and BP

RDA: Range Doppler Algorithm, CSA: Chirp Scaling Algorithm, BP: Back-Projection

One-Step Sparse Image Refinement

- Image-domain regularization [4]

$$\hat{\mathbf{H}} = \arg \min_{\mathbf{H}} \underbrace{\frac{1}{2} \|\mathbf{H} - \mathbf{H}_{\text{CSA}}\|_F^2}_{\text{Preserve CSA image}} + \lambda \underbrace{\|\mathbf{H}\|_1}_{\text{Promote sparsity}}$$

- Strong scatterers $\rightarrow$ retained; weak components $\rightarrow$ suppressed
- $\hat{\mathbf{H}} = \mathcal{S}_{\lambda}(\mathbf{H}_{\text{CSA}})$, $\mathcal{S}_{\lambda}(x) = \frac{x}{|x|} \max(|x| - \lambda, 0)$, $x \neq 0$
- One-Step Refinement
 - $\mathbf{Y} \xrightarrow{\text{CSA}} \mathbf{H}_{\text{CSA}} \xrightarrow{\mathcal{S}_{\lambda}(\cdot)} \hat{\mathbf{H}}$

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Simulation Setup

- ▶ Dataset: 100 offshore SAR images sampled from 1,593 images
- ▶ Evaluation: Ship detection on reconstructed SAR images
- ▶ SAR Simulation Parameters

Parameter	Value	Parameter	Value
Pulse Width	12.5 μ s	Wavelength	0.1152 m
Pulse Repetition Frequency	1 kHz	Closest Slant Range	4000 m
Bandwidth	50 MHz	Platform Velocity	120 m/s
Sampling Frequency	64 MHz	Antenna Length	1.2 m

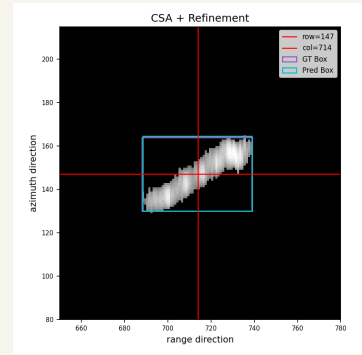
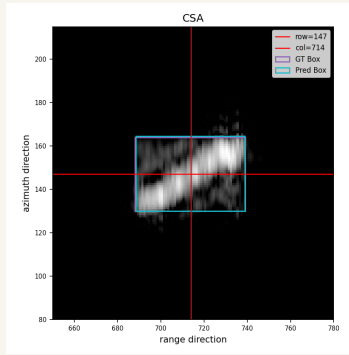
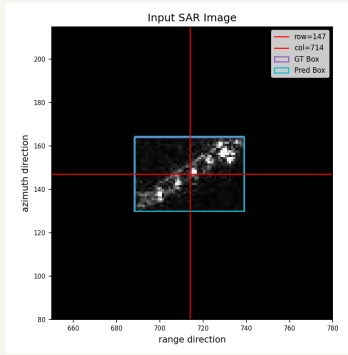
Ship Detection Performance

Image	Precision	Recall	mAP@0.5
Input SAR Image	1.000	0.978	0.975
CSA	0.977	0.940	0.954
CSA + Refinement	0.993	0.986	0.985

- Refinement threshold set to 120 (intensity range: 0–255)
- Thresholding suppresses CSA-induced background artifacts and improves ship detection performance.

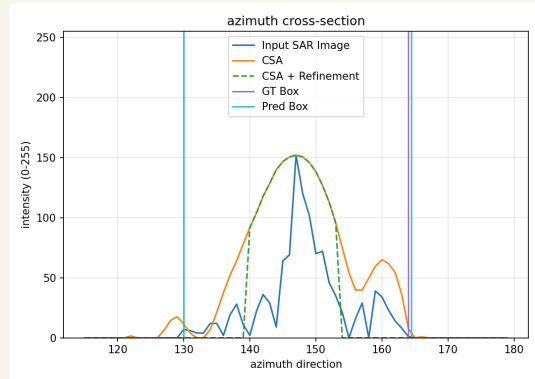
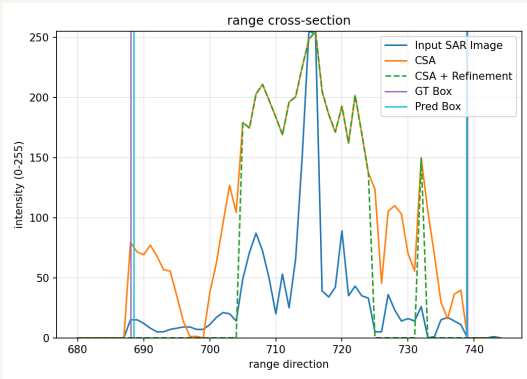
Representative Example

- Observation: Thresholding suppresses background artifacts while preserving the dominant ship response.



Cross-Section Analysis

- Thresholding removes low-amplitude sidelobe/background responses while retaining the dominant target structure.



Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Conclusion

- ▶ Established an end-to-end framework for evaluating SAR image enhancement through ship detection.
- ▶ One-step sparse refinement suppresses background artifacts while preserving dominant ship responses.
- ▶ CSA + refinement improves detection performance from 0.954 to 0.985 mAP@0.5.

Outline

- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
- ▶ Reference

Reference

- [1] F. Song, S. Yang and X. Zhou, "SARES-DEIM: Sparse Mixture-of-Experts Meets DETR for Robust SAR Ship Detection," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS)*, 2026.
- [2] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su and J. Shi, "HRSID: A High-Resolution SAR Images Dataset for Ship Detection and Instance Segmentation," in *IEEE Access*, vol. 8, pp. 120234–120254, 2020.
- [3] T. Zhang *et al.*, "Balance Scene Learning Mechanism for Offshore and Inshore Ship Detection in SAR Images," in *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [4] H. Bi, G. Bi, B. Zhang, W. Hong and Y. Wu, "From Theory to Application: Real-Time Sparse SAR Imaging," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 4, pp. 2928–2936, April 2020.